PHASE 1 - Installing the libraries

```
%pip install -q --user --upgrade google-cloud-aiplatform vertexai
```

Show hidden output

```
import IPython

app=IPython.Application.instance()
app.kernel.do_shutdown(True)
```

```
{'status': 'ok', 'restart': True}
```

```
%pip install -q --upgrade langchain
%pip install -q --upgrade langchain-core
%pip install -q --upgrade langchain-community langchain-google-vertexai
%pip install -q --upgrade --quiet langchain-google-community[gcs]
```

```
import IPython

app=IPython.Application.instance()
app.kernel.do_shutdown(True)
```

```
{'status': 'ok', 'restart': True}
```

```
# intall : utilities libraires needed for Q&A
! sudo apt -y -qq install tesseract-ocr libtesseract-dev
! sudo apt-get -y -11 install poppler-utils

%pip install -q --user --upgrade unstructured pdf2image pytesseract pdfminer.six
%pip install -q --user --upgrade pillow-heif opencv-python unstructured-inference pikepdf pypdf

%pip install -q --user --upgrade tensorflow_hub tensorflow_text

%pip install -q --user --upgrade pi-heif

%pip install -q --user unstructured[inference]
```

Show hidden output

```
import IPython

app=IPython.Application.instance()
app.kernel.do_shutdown(True)
```

```
{'status': 'ok', 'restart': True}
```

```
!pip uninstall torch torchvision torchaudio

!pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118
```

Show hidden output

```
import IPython

app=IPython.Application.instance()
app.kernel.do_shutdown(True)
```

```
{'status': 'ok', 'restart': True}
```

PHASE 2

Import all necessary software application platforms, tool libraries, and utility software applications needed to develop the code for the Q&A Search system

```
from posixpath import splitext
```

```
import sys

if 'google.colab' in sys.modules:
  from google.colab import auth

  auth.authenticate_user()
```

```
from google.cloud import aiplatform
import vertexai

from google.cloud.aiplatform.matching_engine.matching_engine_index_endpoint import (
    Namespace,
    NumericNamespace,
)
```

```
import langchain

from langchain.chains import RetrievalQA
from langchain.prompts import PromptTemplate
from langchain.text_splitter import RecursiveCharacterTextSplitter

from langchain_community.document_loaders import PyPDFLoader
```

```
from langchain_google_community import GCSDirectoryLoader, GCSFileLoader
```

```
#import Fomr GCP: Vertex AI and Langchain API

from langchain_google_vertexai import VertexAI , VertexAIEmbeddings
from langchain_google_vertexai import (
    VectorSearchVectorStore,
    VectorSearchVectorStoreDatastore,
)
```

```
import textwrap

import pi_heif
```

PHASE 3:

Develop the code (COLAB) to initialize the GCP: AI Platform for the Q&A Search system

```
import os

PROJECT_ID = "ivory-infusion-456900-h4"  # @param {type: "st
if not PROJECT_ID or PROJECT_ID == "[your-project-id]": PROJ
REGION = "us-central1"  # @param {type: "string", placeholde
if not REGION or REGION == "[your-region]":  REGION = "us-ce
BUCKET_NAME = "adta5770-docs-folder-g6"  # @param {type: "st
if not BUCKET_NAME or BUCKET_NAME == "[your-bucket-name]": B
BUCKET_URL = f"gs://{BUCKET_NAME}"
PREFIX = "documents/pdfs"
if not PREFIX or PREFIX == "[your-PREFIX]": PREFIX = "[your-
PREFIX = "documents/pdfs"
```

PROJECT_ID

ivory-infusion-456900-h4

REGION

us-central1

BUCKET_NAME

adta5770-docs-folder-g6

Hide code

```
print(f"PROJECT_ID: {PROJECT_ID}")
print(f"REGION: {REGION}")
print(f"BUCKET_NAME: {BUCKET_NAME}")
print(f"BUCKET_URL: {BUCKET_URL}")
print(f"PREFIX: {PREFIX}")
```

```
PROJECT_ID: ivory-infusion-456900-h4
REGION: us-central1
BUCKET_NAME: adta5770-docs-folder-g6
BUCKET_URL: gs://adta5770-docs-folder-g6
PREFIX: documents/pdfs
```

```
# Initialize Vertex AI
aiplatform.init(
```

```
    project=PROJECT_ID
    , location=REGION
    , staging_bucket=BUCKET_URL)
```

PHASE 4:

Create an empty vector search index and deploy it with a public endpoint

```
#---- PHASE 4: STEP 1: Verify the list of currently existing indexes and endpoints
# NOTES: Either none or some already existing.

# First, get the list of indexes that have been created for the vector search system

list_indexes = aiplatform.MatchingEngineIndex.list()

print(f"List of indexes: {list_indexes}")
```

```
List of indexes: []
```

```
# Next, get the list of end-points that have been created

list_end_points = aiplatform.MatchingEngineIndexEndpoint.list()

print(f"List of indexes: {list_end_points}")
```

```
List of indexes: []
```

```
#---- PHASE 4: STEP 2: DELETE ALL currently existing indexes and endpoints
# SEE THE CODE OF HOW TO DELETE ALL EXISTING INDEXES AND ENDPOINTS IN ANOTHER DOCUMENT
# NOTE: Not executed because i have empty indexes
del_index_1 = aiplatform.MatchingEngineIndex("9082847853755236352")
del_index_1.delete()
```

```
del_index_endpoint_1 = aiplatform.MatchingEngineIndexEndpoint("5114121050879164416")
del_index_endpoint_1.undeploy_all()
del_index_endpoint_1.delete()
```

```
#---- PHASE 4: STEP 3: VERIFY that NO  index and endpoint currently exist

list_indexes = aiplatform.MatchingEngineIndex.list()

print(f"List of indexes: {list_indexes}")
```

```
List of indexes: []
```

```
list_end_points = aiplatform.MatchingEngineIndexEndpoint.list()

print(f"List of indexes: {list_end_points}")
```

```
List of indexes: []
```

```
# NOTES: Run the code and nothing is printed out --> The list is empty: GOOD!
```

```
#---- PHASE 4: STEP 4: Create an empty index

# Define constant identifiers: Text names of index and endpoint

# The number of dimensions for the textembedding-005 is 768
DIMENSIONS = 768

A_XYZ_DISPLAY_INDEX_NAME = "Andy_G6"
A_XYZ_DISPLAY_END_POINT_NAME = "Andy_End_Point_G6"
A_XYZ_DEPLOYED_INDEX_ID = "Andy_INDEX_ID_G5"
```

```
#---- PHASE 4: STEP 5: Define text embedding model
embedding_model = VertexAIEmbeddings(model_name="text-embedding-005")
```

```
#---- PHASE 4: STEP 6: Create a new empty vector search index and verify it has been created successfully
a_new_index = aiplatform.MatchingEngineIndex.create_tree_ah_index(
    display_name=A_XYZ_DISPLAY_INDEX_NAME,
```

```
    dimensions=DIMENSIONS,
    approximate_neighbors_count=150,
    distance_measure_type="DOT_PRODUCT_DISTANCE",
    index_update_method="STREAM_UPDATE"
)

if a_new_index:
    print(a_new_index.name)

list_indexes = aiplatform.MatchingEngineIndex.list()

print(list_indexes)
```

```
217511987276414976
[<google.cloud.aiplatform.matching_engine.matching_engine_index.MatchingEngineIndex object at 0x7982e5f40450>
resource name: projects/87882215990/locations/us-central1/indexes/217511987276414976]
```

```
#---- PHASE 4: STEP 7: Create a new index ednpoint and verify it has been created successfully
a_new_endpoint = aiplatform.MatchingEngineIndexEndpoint.create(
    display_name=A_XYZ_DISPLAY_END_POINT_NAME,
    public_endpoint_enabled=True
)

if a_new_endpoint:
    print(a_new_endpoint.name)

list_end_points = aiplatform.MatchingEngineIndexEndpoint.list()

print(list_end_points)
```

```
3150551613345628160
[<google.cloud.aiplatform.matching_engine.matching_engine_index_endpoint.MatchingEngineIndexEndpoint object at 0x7982ec157210>
resource name: projects/87882215990/locations/us-central1/indexEndpoints/3150551613345628160]
```

```
#---- PHASE 4: STEP 8: Deploy the newly created index with a public endpoint

a_new_endpoint = a_new_endpoint.deploy_index(
    index=a_new_index,
    deployed_index_id=A_XYZ_DEPLOYED_INDEX_ID
)

a_new_endpoint.deployed_indexes
```

```
[id: "Andy_INDEX_ID_G5"
index: "projects/87882215990/locations/us-central1/indexes/217511987276414976"
create_time {
  seconds: 1745958473
  nanos: 999681000
}
index_sync_time {
  seconds: 1745960134
  nanos: 788686000
}
automatic_resources {
  min_replica_count: 2
  max_replica_count: 2
}
deployment_group: "default"
]
```

```
###======================== AT THIS POINT:
# A new empty vector search index has been successfully created.
# A new index end point has been successfully created.
# The newly created index has been deployed with a public end point successfully

# COMPLETE SEMESTER PROJECT CODE: PHASE 4 --> READY FOR PHASE 5: EMDBEDDING PDF FILES
```

PHASE 5

```
from google.cloud import storage
from langchain.document_loaders import PyPDFLoader  # If PDFs
from langchain.text_splitter import RecursiveCharacterTextSplitter

client = storage.Client()
bucket = client.get_bucket(BUCKET_NAME)
blobs = bucket.list_blobs(prefix=PREFIX)
```

```
from langchain_google_community import GCSDirectoryLoader
from langchain_google_community import GCSFileLoader
```

PHASE 6

```
all_documents = []
```

```
import warnings
warnings.filterwarnings('ignore', message='.*CropBox missing from /Page.*')
```

```
from google.cloud import storage
client = storage.Client()
for blob in client.list_blobs(BUCKET_NAME, prefix="documents/pdfs"):
  print(str(blob))
```

```
<Blob: adta5770-docs-folder-g6, documents/pdfs/, 1745894724712362>
<Blob: adta5770-docs-folder-g6, documents/pdfs/TSLA-Q2-2021-Update.pdf, 1745900085465805>
<Blob: adta5770-docs-folder-g6, documents/pdfs/TSLA-Update-Letter-2018-2Q.pdf, 1745900085965309>
<Blob: adta5770-docs-folder-g6, documents/pdfs/TSLA_Update_Letter_2019-1Q.pdf, 1745900083655776>
<Blob: adta5770-docs-folder-g6, documents/pdfs/TSLA_Update_Letter_2019-2Q.pdf, 1745900083638451>
<Blob: adta5770-docs-folder-g6, documents/pdfs/Team-5-Executive-Summary-Tesla.pdf, 1745900078891584>
<Blob: adta5770-docs-folder-g6, documents/pdfs/Tesla-Motors-Inc..pdf, 1745900081658705>
<Blob: adta5770-docs-folder-g6, documents/pdfs/Tesla.pdf, 1745900079297648>
<Blob: adta5770-docs-folder-g6, documents/pdfs/Tesla2024_report.pdf, 1745940300993434>
<Blob: adta5770-docs-folder-g6, documents/pdfs/Tesla_Motors_Q1_15_Shareholder_Letter.pdf, 1745900079764429>
<Blob: adta5770-docs-folder-g6, documents/pdfs/modern-electric-cars-of-tesla-motors-company-3kqxwjq51o.pdf, 1745900074600355>
<Blob: adta5770-docs-folder-g6, documents/pdfs/nasdaq-tsla-2024-10K-24569853.pdf, 1745900076629131>
<Blob: adta5770-docs-folder-g6, documents/pdfs/ssrn-3896901.pdf, 1745900078140576>
<Blob: adta5770-docs-folder-g6, documents/pdfs/tesla model.pdf, 1745900079101665>
<Blob: adta5770-docs-folder-g6, documents/pdfs/tesla2017.pdf, 1745900080240706>
<Blob: adta5770-docs-folder-g6, documents/pdfs/tsla-20240331-gen.pdf, 1745900084751333>
```

```
# Make sure these variables are set in your environment or defined here
PROJECT_ID = PROJECT_ID  # Replace with your Project ID if not using env var
GCS_BUCKET_NAME = BUCKET_NAME   # Replace with your bucket name (e.g., GCS_BUCKET_DOCS)
FOLDER_PREFIX = PREFIX          # Replace with folder prefix in the bucket (folder_prefix)
# Ensure FOLDER_PREFIX ends with a '/' if it's targeting a directory structure

print(f"Processing documents from gs://{GCS_BUCKET_NAME}/{FOLDER_PREFIX}")
storage_client = storage.Client(project=PROJECT_ID)
bucket = storage_client.bucket(GCS_BUCKET_NAME)

all_documents = []
blobs = bucket.list_blobs(prefix=FOLDER_PREFIX)  # List blobs matching the prefix

for blob in blobs:
  print(str(blob))
  # Skip directories/folders if represented as blobs, and ensure it's a PDF
  if blob.name.endswith("/") or not blob.name.lower().endswith(".pdf"):
      continue

  print(f" Loading document: {blob.name}")

  loader = GCSFileLoader(
      project_name=PROJECT_ID, bucket=GCS_BUCKET_NAME, blob=blob.name
  )
  # Load documents (GCSFileLoader often returns one document per page)
  documents_from_blob = loader.load()

  #=================== NEW NEW NEW ===================

  # --- Metadata Enhancement (similar to original logic) ---
  # Derive document name from the blob name
  document_name = blob.name.split("/")[-1]

  # Derive doc source prefix and suffix to match original logic
```

```
    # Note: The original code's 'source' derivation seemed to point to the folder rather than the file. This replicates that.
    #       A more common approach might be to use the blob's full gs:// path as the source.
    doc_source_prefix = f"gs://{GCS_BUCKET_NAME}"
    # Get the directory path containing the blob
    doc_source_suffix = "/".join(blob.name.split("/")[:-1])
    source = f"{doc_source_prefix}/{doc_source_suffix}"  # Folder path as source

    for document in documents_from_blob:
      # Add derived metadata to each document (page)
      document.metadata["source"] = source
      document.metadata["document_name"] = document_name
      # GCSFileLoader might add other useful metadata like 'page' automatically
      all_documents.extend(documents_from_blob)

# The 'all_documents' list now contains LangChain Document objects,
# likely one per page from all loaded PDFs.
print(f"# of document pages loaded (pre-chunking) = {len(all_documents)}")
```

Show hidden output

```
len(all_documents[:50])
```

```
15
```

```
text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=1000,
    chunk_overlap=50,
    length_function=len,
    is_separator_regex=False,
)

doc_splits= text_splitter.split_documents(all_documents)
for idx, split in enumerate(doc_splits):
    split.metadata["chunk"]= idx

print(f"# of documents = {len(doc_splits)}")
```

```
# of documents = 1159
```

PHASE 7

Define Cred for Vector store

```
#run
# Define Cred for Vector store
VSVDB_REGION = "us-central1"
VSVDB_INDEX_NAME = "g6-vsvd-index"
VSVDB_EMBEDDING_DIR = "g6-vsvd-bbucket"
VSVDB_DIMENSIONS = 768
```

```
#create GCS bucket to store the matching Engine/vector serch index
! set -x && gsutil mb -p $PROJECT_ID -l us-central1 gs://$VSVDB_EMBEDDING_DIR
```

```
+ gsutil mb -p ivory-infusion-456900-h4 -l us-central1 gs://g6-vsvd-bbucket
Creating gs://g6-vsvd-bbucket/...
```

```
#run
embeddings = VertexAIEmbeddings(model= "text-embedding-005")
```

```
list_indexes = aiplatform.MatchingEngineIndex.list()
index_endpoints = aiplatform.MatchingEngineIndexEndpoint.list()
for indexes in list_indexes:
  print(f"indexes: {indexes.name}")
for endpoint in index_endpoints:
  print(f"Endpoint: {endpoint.name}")
for deployed_index in a_new_endpoint.deployed_indexes:
  print(f"Deployed Index: {deployed_index.id}")
```

```
indexes: 217511987276414976
Endpoint: 3150551613345628160
Deployed Index: Andy_INDEX_ID_G5
```

```
#run
```

```
index = aiplatform.MatchingEngineIndex(index_name='2175119872764149 76')
index_endpoint = aiplatform.MatchingEngineIndexEndpoint(index_endpoint_name='3150551613345628160')
```

```
#run
vsvectordb = VectorSearchVectorStore.from_components(
    project_id = PROJECT_ID,
    region = VSVDB_REGION,
    gcs_bucket_name =f"gs://{VSVDB_EMBEDDING_DIR}".split("/")[2],
    index_id = index.name,
    embedding = embeddings,
    #documents = doc_splits,
    endpoint_id=index_endpoint.name,
    stream_update=True,
)
```

```
texts = [doc.page_content for doc in doc_splits]
metadata = [doc.metadata for doc in doc_splits]
```

```
print(f"Number of texts: {len(texts)}")
print(f"Number of metadata: {len(metadata)}")
```

```
Number of texts: 1159
Number of metadata: 1159
```

Double-click (or enter) to edit

```
# Add embeddings to the Vector store
for i in range(0, len(texts),1000):
  vsvectordb.add_texts(texts[i:i+1000], metadata[i:i+1000])
```

PHASE 8 (RUN)

```
vsvectordb.similarity_search("What are the risk factors of heart attack in pregnant woman?",k=2)
```

```
[Document(id='caea4325-f886-467c-bcb5-5ad3d9bac88b', metadata={'source': 'gs://adta5770-docs-folder-g6/documents/pdfs',
'document_name': 'Tesla.pdf', 'chunk': 287}, page_content='20'),
 Document(id='e5dcfe65-f183-4766-a180-2db7a26b3e4f', metadata={'source': 'gs://adta5770-docs-folder-g6/documents/pdfs',
'document_name': 'Tesla.pdf', 'chunk': 281}, page_content='19')]
```

```
vsvectordb.similarity_search("Why people with lung disease are susceptible to death in Covid-19?",k=2)
```

```
[Document(id='9ac364df-400e-4f3f-8c14-09d83f809fa7', metadata={'source': 'gs://adta5770-docs-folder-g6/documents/pdfs',
'document_name': 'nasdaq-tsla-2024-10K-24569853.pdf', 'chunk': 492}, page_content='our business may suffer if we cannot
effectively manage the resulting greater levels of residual risk.'),
 Document(id='aa5d8408-9c24-4d95-b40e-a8b0eb1f3367', metadata={'source': 'gs://adta5770-docs-folder-g6/documents/pdfs',
'document_name': 'nasdaq-tsla-2024-10K-24569853.pdf', 'chunk': 533}, page_content='We may be impacted by natural disasters,
wars, health epidemics, weather conditions, the long-term effects of climate change, power outages or')]
```

PHASE 9

```
from langchain_google_vertexai import VertexAI

llm = VertexAI (
    model_name = "gemini-2.5-pro-exp-03-25",
    max_output_tokens = 8192,
    temperature = 0.2,
    top_p = 0.8,
    top_k = 40,
    verbose = True
)
```

```
NUMBER_OF_RESULTS = 10

SEARCH_DISTANCE_THRESHOLD = 0.6

retriever = vsvectordb.as_retriever(
    search_type = "similarity",
    search_kwargs = {
        "k": NUMBER_OF_RESULTS,
        "search_distance": SEARCH_DISTANCE_THRESHOLD
    },
```

```
    filters= None,
)
```

```
template = """
          **Context:** You are a professional medical advisor. Your task is to critically analyze a particular information from a

          **User Query:**
          "{question}"

          Strictly use only the following pieces of context to answer the question below. Think step by step.
          Do not try to make up an answer:
                          If the context does not contain information or knowledge about the user's/patient's query, just say tha
                          If the context is empty, just say that you are out of answer and you don't know answer — do not attempt

          **Context:**
          ---
          {context}
          ---

          **Analysis Instructions:**
          1.  **Understand the Query:** What specific information is the user seeking? (e.g., symptoms, treatments, risk factors,
          2.  **Scan the Excerpt:** Carefully read the provided medical or health excerpt.
          3.  **Identify Relevance:** Highlight statements, data points, or explanations within the excerpt that *directly* addre
          4.  **Synthesize Findings:** Using *only* the relevant information identified in the excerpt, create a clear and accura
              *   Aim for 3-5 concise sentences.
              *   If the query seeks details (like side effects, statistics, or mechanisms), include them where available.
              *   Ensure the summary stays focused on directly answering the user's query.
          5.  **Handle Irrelevance:** If the excerpt does not contain any information relevant to the query, clearly state: "The

          **Synthesized Summary:**
"""
```

```
#from re import VERBOSE
reteriveal_qa = RetrievalQA.from_chain_type(
    llm = llm,
    chain_type = "stuff",
    retriever = retriever,
    return_source_documents = True,
    verbose = True,
    chain_type_kwargs = {
        "prompt": PromptTemplate
        (
                        template = template,
            input_variables = ["context","question"],
        ),
    },
)
```

```
# enable for troublshooting
reteriveal_qa.combine_documents_chain.verbose = True
reteriveal_qa.combine_documents_chain.llm_chain.verbose = True
reteriveal_qa.combine_documents_chain.llm_chain.llm.verbose = True
```

PHASE 10

```
def formatter(result):
  print(f"questions: {result['query']}")
  print("."*80)
  print(f"Answer: {result['result']}")
  print(f"Sources: {result['source_documents']}")
  if "source_documents" in result.keys():
    for idx,ref in enumerate(result["source_documents"]):
      print("."*80)
      print(f"Reference #: {idx}")
      print("."*80)
      if "score" in ref.metadata:
        print(f"Matching Score: {ref.metadata['score']}")
      if "source" in ref.metadata:
        print(f"Source: {ref.metadata['source']}")
      if "document_name" in ref.metadata:
        print(f"Document Name: {ref.metadata['document_name']}")
```

```
        print("."*80)
        print(f"Content: \n{ref.page_content}")
    print("."*80)
    print(f"Response:{wrap(result['result'])}")
    print("."*80)

def wrap(s):
    return "\n".join(textwrap.wrap(s, width=120, break_long_words=False))


def ask(
      query,
      qa=reteriveal_qa,
      k=NUMBER_OF_RESULTS,
      search_distance_threshold=SEARCH_DISTANCE_THRESHOLD,
      filters={}
):
    reteriveal_qa.retriever.search_kwargs['search_distance'] = search_distance_threshold
    reteriveal_qa.retriever.search_kwargs['k'] = k
    reteriveal_qa.retriever.search_kwargs['filters'] = filters
    result = reteriveal_qa({"query":query})
    formatter(result)
```

```
ask("What is Tesla?")
```

```


> Entering new RetrievalQA chain...


> Entering new StuffDocumentsChain chain...


> Entering new LLMChain chain...
Prompt after formatting:

          **Context:** You are a professional medical advisor. Your task is to critically analyze a particular information fro

          **User Query:**
          "What is Tesla?"

          Strictly use only the following pieces of context to answer the question below. Think step by step.
          Do not try to make up an answer:
                          If the context does not contain information or knowledge about the user's/patient's query, just say
                          If the context is empty, just say that you are out of answer and you don't know answer — do not atte

          **Context:**
          ---
          Abstract

This article is about Tesla, Inc., which was founded in 2003 by a group of engineers in Silicon Valley whose objective was to

Introduction

Abstract. Incorporated in 2003, Tesla motors, inc is a leading American automotive company named after electrical engineer and

Building on the success of its sedan, Tesla introduced the Model X in 2015, which became the

safest SUV ever tested. It earned a 5-star safety ratings across every category from the National

Highway Traffic Safety Administration, the first vehicle to ever hit that mark. That same year,

Tesla began rolling out an auto-pilot feature for partial autonomous driving, which enabled the

cars to self-drive with active driver supervision.

With three successful vehicle launches under its belt, Tesla began to focus on the energy side

of its business, unveiling Solar Roof solar tiles as well as Powerwall and Powerpack industry

batteries to store converted solar energy. To support this focus on energy, Tesla strategically

acquired SolarCity, a solar energy service company, in 2016 for $2.6 billion. To seal the deal, the

company changed its name from Tesla Motors to Tesla, Inc. in 2017, signifying the company's

shift from a car manufacturer to a sustainable energy solution company.

Tesla Factory in Fremont, CA, just 20 miles away from Mr. Musk's office. Also notable is the
```

```
Tesla Gigafactory, still under construction, a 2 million sq. ft. facility near Reno, NV, a joint-
```

```
ask("What is the current stock price of TSLA?")
```

```


> Entering new RetrievalQA chain...


> Entering new StuffDocumentsChain chain...


> Entering new LLMChain chain...
Prompt after formatting:

          **Context:** You are a professional medical advisor. Your task is to critically analyze a particular information fro

          **User Query:**
          "What is the current stock price of TSLA?"

          Strictly use only the following pieces of context to answer the question below. Think step by step.
          Do not try to make up an answer:
                          If the context does not contain information or knowledge about the user's/patient's query, just say t
                          If the context is empty, just say that you are out of answer and you don't know answer — do not atte

          **Context:**
          ---
          5-YR AVERAGE

19.69

10.13 42.31

Company Outlook The company anticipates a lower vehicle volume growth rate in 2024 due to a focus on launching a new vehicle p

Dividend Yield

0.0 %

REVENUE - %CHG YOY

1Q23

2Q23

3Q23

4Q23

1Q24E

Automotive

20.2% 45.7% 7.1%

3.1%

5.6%

Services & Other Energy Generation & Storage
```

```
ask("What was Tesla revenue last quarter?")
```

```


> Entering new RetrievalQA chain...


> Entering new StuffDocumentsChain chain...


> Entering new LLMChain chain...
Prompt after formatting:

          **Context:** You are a professional medical advisor. Your task is to critically analyze a particular information fro

          **User Query:**
          "What was Tesla revenue last quarter?"

          Strictly use only the following pieces of context to answer the question below. Think step by step.
```

```
          Do not try to make up an answer:
                         If the context does not contain information or knowledge about the user's/patient's query, just say t
                         If the context is empty, just say that you are out of answer and you don't know answer — do not atter

          **Context:**
          ---
          Revenue growth of each business In the fourth quarter, Tesla posted revenue of $25.17 billion, slightly below the exp

SENTIMENT 1Q23

2Q23

3Q23

4Q23

net profit margin revenue

Positive Positive Negative Positive Negative Negative Negative Negative Positive Positive Positive Positive

12,696

3,508

13,626

3,388

49,616

5,989

5,229

29,725

4,180

184
```

```
ask("Summarize recent news about Tesla?")
```

```


> Entering new RetrievalQA chain...


> Entering new StuffDocumentsChain chain...


> Entering new LLMChain chain...
Prompt after formatting:

          **Context:** You are a professional medical advisor. Your task is to critically analyze a particular information fror

          **User Query:**
          "Summarize recent news about Tesla?"

          Strictly use only the following pieces of context to answer the question below. Think step by step.
          Do not try to make up an answer:
                         If the context does not contain information or knowledge about the user's/patient's query, just say t
                         If the context is empty, just say that you are out of answer and you don't know answer — do not atter

          **Context:**
          ---
          the allegations.

In addition to the whistleblower scandal, it was discovered that Tesla violated labor laws. In a

court ruling on September 27, 2019, Tesla was found guilty of violating the National Labor

Relations Act several times, which included specific counts of threatening employees and

retaliating against them. This court case was tried in California for incidents that happened in 2017.

Also, Musk tweeted in May 2018 that joining a union meant giving up Tesla stock options, which

was also illegal, the judge found. The company fired a union supporter in the past, and the judge

stated that this person must be reinstated and given back pay for wrongful termination. There is

also a notice that Tesla must announce to its employees that the company violated labor laws.
```

```
Tesla can appeal this ruling which seems likely. This string of ethical issues could indicate deeper

issues at Tesla.

STRUGGLING SUPPLY CHAIN

success, Tesla and Musk are now household names not only in the United States but also around

the world. With more recognition and awareness from consumers and investors comes more

scrutiny. Tesla, which has fumbled when it comes to setting expectations and meeting deadlines,

will need to streamline its operations if it wants to be a contender in the mass car market.

Additionally, Tesla will need to face its ethical misconduct head on to prevent wrongdoing from
```

```
ask("What is the trading volume for tesla today?")
```

```
> Entering new RetrievalQA chain...


> Entering new StuffDocumentsChain chain...


> Entering new LLMChain chain...
Prompt after formatting:

          **Context:** You are a professional medical advisor. Your task is to critically analyze a particular information fro

          **User Query:**
          "What is the trading volume for tesla today?"

          Strictly use only the following pieces of context to answer the question below. Think step by step.
          Do not try to make up an answer:
                         If the context does not contain information or knowledge about the user's/patient's query, just say
                         If the context is empty, just say that you are out of answer and you don't know answer — do not atte

          **Context:**
          ---
          3,248,827 $ 2,276,951 762,810 480,225 768,276 1,705,711 9,242,800 9,787,950 1,157,343 211,390 2,475,135 22,874,618 5

3,404,451 2,094,253 630,292 502,840 792,601 2,567,699 9,992,136 9,403,672 990,873 328,926 2,710,403 23,426,010 555,964 4,923,2

6,517,022 $ 3,485,597 $

7,080,584 3,551,891

Tesla, Inc. Condensed Consolidated Statement of Cash Flows (Unaudited) (In thousands)

386,129 664,072 750,759 1,351,190 160,816 103,434 1,087,574 1,240,322 2,176,078 (689,289) 19,124 (329,432) (15,006) (728,999 )

340,174 703,929 43,471

(521,831) 8,762 (157,453) 25,750 (644,772) 22,873 (667,645)

(621,392 ) 5,064 (163,582 ) 50,911

13,707

753,225 1,437,163 103,434 2,293,822 (1,218,366) 10,278 (313,128) 13,195 (1,508,021) 19,312 (1,527,333)

19,072

34,490

(25,167 )

53,562

(100,243)

$
```

```
ask("What is the effect of a broker non-vote?")
```

```
> Entering new RetrievalQA chain...


> Entering new StuffDocumentsChain chain...


> Entering new LLMChain chain...
Prompt after formatting:

          **Context:** You are a professional medical advisor. Your task is to critically analyze a particular information fro

          **User Query:**
          "What is the effect of a broker non-vote?"

          Strictly use only the following pieces of context to answer the question below. Think step by step.
          Do not try to make up an answer:
                          If the context does not contain information or knowledge about the user's/patient's query, just say
                          If the context is empty, just say that you are out of answer and you don't know answer — do not atte

          **Context:**
          ---
          will be based on the rating assigned to our senior, unsecured long-term indebtedness (the "Credit Rating") from time

our business may suffer if we cannot effectively manage the resulting greater levels of residual risk.

a material impact on our consolidated financial statements.

unreasonable result. Tesla believes that the adjustments made by JP Morgan were neither proper nor commercially reasonable, as

known    would    lead    to      a       commercially    unreasonable    result. Tesla   believes       that    the    adjust

to pursue an enforcement action, there exists the possibility of a material adverse impact on our business, results of operati

We may fail to meet our publicly announced guidance or other expectations about our business, which could cause our stock pric

statement       which    included       a       number   of      proposals,       including       a       proposal        to

Noncontrolling interests and redeemable noncontrolling interests represent third-party interests in the net assets under certa

contracts and cause us to cease solar and energy storage system sales and services in the relevant jurisdictions, which may ha
          ---

          **Analysis Instructions:**
          1.  **Understand the Query:** What specific information is the user seeking? (e.g., symptoms, treatments, risk facto
          2.  **Scan the Excerpt:** Carefully read the provided medical or health excerpt.
          3.  **Identify Relevance:** Highlight statements, data points, or explanations within the excerpt that *directly* ad
          4.  **Synthesize Findings:** Using *only* the relevant information identified in the excerpt, create a clear and acc
              *   Aim for 3-5 concise sentences.
              *   If the query seeks details (like side effects, statistics, or mechanisms), include them where available.
              *   Ensure the summary stays focused on directly answering the user's query.
          5.  **Handle Irrelevance:** If the excerpt does not contain any information relevant to the query, clearly state: "T

          **Synthesized Summary:**
```

```
ask("How can i buy stock?")
```

```
> Entering new RetrievalQA chain...


> Entering new StuffDocumentsChain chain...


> Entering new LLMChain chain...
Prompt after formatting:

          **Context:** You are a professional medical advisor. Your task is to critically analyze a particular information fro

          **User Query:**
          "How can i buy stock?"

          Strictly use only the following pieces of context to answer the question below. Think step by step.
          Do not try to make up an answer:
                          If the context does not contain information or knowledge about the user's/patient's query, just say
                          If the context is empty, just say that you are out of answer and you don't know answer — do not atte

          **Context:**
          ---
          public offering was priced at approximately $1.13 per share on June 28, 2010 as adjusted to give effect to the three

Holders
```

```
As of January 22, 2024, there were 9,300 holders of record of our common stock. A substantially greater number of holders of ou

are “street name” or beneficial holders, whose shares are held by banks, brokers and other financial institutions.

Dividend Policy

We have never declared or paid cash dividends on our common stock. We currently do not anticipate paying any cash dividends in

17,576 345 223 18,144

Investments

Investments may be comprised of a combination of marketable securities, including U.S. government securities, corporate debt se

The cost of available-for-sale investments sold is based on the specific identification method. Realized gains and losses on th

While we intend to continue to leverage our most effective sales strategies, including sales through our website, we may not be

91-2197729 (I.R.S. Employer Identification No.)

1 Tesla Road Austin, Texas (Address of principal executive offices)

78725 (Zip Code)

(512) 516-8177 (Registrant’s telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class Common stock
```

```
ask("What is the short interest in TSLA stock?")
```

```
> Entering new RetrievalQA chain...


> Entering new StuffDocumentsChain chain...


> Entering new LLMChain chain...
Prompt after formatting:

          **Context:** You are a professional medical advisor. Your task is to critically analyze a particular information fro

          **User Query:**
          "What is the short interest in TSLA stock?"

          Strictly use only the following pieces of context to answer the question below. Think step by step.
          Do not try to make up an answer:
                          If the context does not contain information or knowledge about the user's/patient's query, just say t
                          If the context is empty, just say that you are out of answer and you don't know answer — do not atter

          **Context:**
          ---
          shares of common stock in certain of our public offerings and private placements at the same prices offered to third

Anti-takeover provisions contained in our governing documents, applicable laws and our convertible senior notes could impair a

386,129 664,072 750,759 1,351,190 160,816 103,434 1,087,574 1,240,322 2,176,078 (689,289) 19,124 (329,432) (15,006) (728,999 )

340,174 703,929 43,471

(521,831) 8,762 (157,453) 25,750 (644,772) 22,873 (667,645)

(621,392 ) 5,064 (163,582 ) 50,911

13,707

753,225 1,437,163 103,434 2,293,822 (1,218,366) 10,278 (313,128) 13,195 (1,508,021) 19,312 (1,527,333)

19,072

34,490

(25,167 )

53,562

(100,243)
```

```
$

(408,334) $

(702,135) $

(717,539 ) $ (1,110,469) $ (1,427,090)
```

```
ask("Has Tesla ever issued dividends, and if so, when was the last time?")
```

```
> Entering new RetrievalQA chain...


> Entering new StuffDocumentsChain chain...


> Entering new LLMChain chain...
Prompt after formatting:

          **Context:** You are a professional medical advisor. Your task is to critically analyze a particular information fro

          **User Query:**
          "Has Tesla ever issued dividends, and if so, when was the last time?"

          Strictly use only the following pieces of context to answer the question below. Think step by step.
          Do not try to make up an answer:
                          If the context does not contain information or knowledge about the user's/patient's query, just say
                          If the context is empty, just say that you are out of answer and you don't know answer — do not atte

          **Context:**
          ---
          future. Any future determination to declare cash dividends will be made at the discretion of our board of directors,

Stock Performance Graph

This performance graph shall not be deemed "filed" for purposes of Section 18 of the Securities Exchange Act of 1934, as amend

The following graph shows a comparison, from January 1, 2019 through December 31, 2023, of the cumulative total return on our

Financing Activities

It is Tesla's main source of cash: the issuing of common stock on the market, added to the low

cost of such issuing, relative low payments on capital leases and other debt (i.e. low exposure to

external lessors and creditors) and not paying dividends contributes to the positive result of

roughly $1.5 Bn. It is a signal that the TSLA ticker is still attractive to investors, an assumption

favorable for the future of the company but insufficient for its present.

E. Dividends policy and stock situation

Tesla Motors has not paid dividends so far and is not likely to do so as long as it does not report

annual profits. However, it is unknown when that will be or if it is willing to revise its dividend

policy at any time soon even if it does manage to make profits. Tesla's stock follows the pattern of

a high growth company. It is a young firm that invests a lot on its future and its stock follows

5-YR AVERAGE

19.69
```

```
ask("Who are the largest institutional holders of TSLA stock?")
```

```
> Entering new RetrievalQA chain...


> Entering new StuffDocumentsChain chain...


> Entering new LLMChain chain...
Prompt after formatting:
```